Using the MLP prior with PhysX and PhysGaussian

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Abstract

The fitted splat is the object. A multilayer perceptron reads each in-mask Gaussian, and the 33 outputs are averaged. That mean vote builds one PhysX actor or a set of PhysGaussian particles. PhysX still owns contact. The weights are epoch 10 of the 14-epoch run in this project. They are not a released third-party checkpoint. The test scores below measure the prior only. The combined system has not been run.

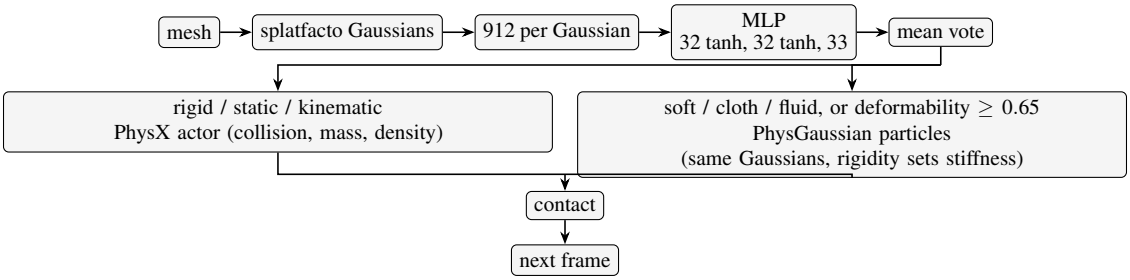

Figure 1: Fit once, from the mesh to the mean vote. The vote branches. Both sides meet at contact, and the next frame follows. This is the plan. The combined system has not been run.

1. Fitted splat

The fitted splat is the object. Each Gaussian is one row of width 912. Columns 0–383 are DINOv2 ViT-S/14, width 384. Columns 384–895 are OpenCLIP ViT-B-16 with the OpenAI weights, width 512. Columns 896–911 are geometry, in this order: position x, y, z, three exp scales, sigmoid opacity, quaternion components `rot_0` through `rot_3`, spherical-harmonic DC coefficients `f_dc_0` through `f_dc_2`, log volume, and anisotropy.

Opacity must vary. A fitted row is kept only when the opacity standard deviation is at least 10^{-3} and the opacity is not the constant 0.99. Constant 0.99 is the wrong input. A triangle prior with opacity fixed at 0.99 is not this model, and the file `physics_prior.npz` is not part of this repository.

2. MLP

The MLP is two tanh layers of width 32 and a linear map to 33. It runs per in-mask Gaussian. The 33 outputs are averaged. Averaging the 912 inputs and running the MLP once is a different function and is not the inference used here. The sigmoid is not inside the forward pass. It is applied to rigidity and deformability after the forward pass, in the loss and when the mean is decoded.

Slots 0–9 are material: plastic, metal, wood, fabric, paper and cardboard, rubber and foam, glass and ceramic, food and organic matter, liquid and granular matter, unknown. Slots 10–15 are body: dynamic rigid, static rigid, kinematic, soft deformable, cloth shell, fluid or granular. Slots 16–23 are collision: box, sphere, cylinder, capsule, convex hull, triangle mesh, tetrahedral mesh, cloth mesh. Slot 24 is log mass and slot 25 is log density, both natural logs. Slot 26 is rigidity and slot 27 is deformability. Slots 28–32 are masked. They receive no target and no gradient. Friction and restitution are unset. Young's modulus, Poisson's ratio, and the rendering material are unpublished and were not invented.

A class tie returns no class. A proposal whose class is unknown has its weight multiplied by 0.4. The keyword proposal is passed in by the caller. The vote does not look up the keyword dictionary on its own.

The saved weights are epoch 10 of the 14-epoch run, file `data/mlp/physics_prior_splat.npz`. The weights were trained in this project. On the test split, $n = 323$, material accuracy is 0.889, body accuracy is 0.991, and collision accuracy is 0.768. These scores measure the prior only, not a PhysX or PhysGaussian rollout. The body accuracy is the body head compared with the stored label. The soft-body rule is applied when a physics record is built, not inside the masked training loss.

3. PhysX

PhysX builds one solid actor from the prior when the body is dynamic rigid, static rigid, or kinematic. The actor takes the collision primitive, the mass, and the density. Friction and restitution stay at the PhysX default, because the prior leaves them unset.

4. PhysGaussian

PhysGaussian is used when the body is soft deformable, cloth shell, or fluid or granular, or when deformability is at least 0.65. An object becomes a soft body when deformability is

at least 0.65, and that rule wins when it fires. The same Gaussians are the particles. MLP rigidity sets stiffness. MLP mass is the particle mass. PhysX still owns contact. That contact loads the particles. The moved Gaussians are then rendered. Both branches meet at contact, and the next frame follows.

5. Frame order

The fit runs once. The orbit is 16 views: 8 azimuths at elevations 18° and 50°. COLMAP registers those views, splatfacto runs for 600 steps, and the pack writes $(N, 912)$. Opacity must vary on that pack. The MLP runs, and the mean vote branches. Each later frame steps PhysX, then steps PhysGaussian under that contact when the object is on the PhysGaussian branch, then renders.

6. Scope

The 3274 / 139 / 323 split is the prior's split. Train has 3274, validation has 139, and test has 323. The test split was scored once and did not choose the epoch. Reproducing the SceneAgent simulation number (Hollis, Fan, and Yang, arXiv:2610.34745, section 3.4) still needs their evaluation scenes and their weights. Those weights are not public. This note is how our prior would drive the two simulators. The combined system has not been run.